Jian Cui

Ph.D. student in Computer Science, University of Illinois Urbana–Champaign
jiancui8682@gmail.com • Google Scholar

EDUCATION

University of Illinois Urbana–Champaign *Aug. 2025 – 2028 (expected)*
Ph.D. in Computer Science. *Advisor:* Xiaojing Liao

Indiana University Bloomington *Aug. 2023 – Aug. 2025*
Ph.D. student in Computer Science. *Advisor:* Xiaojing Liao

Korea Advanced Institute of Science and Technology (KAIST) *Mar. 2015 – Feb. 2022*
B.S. & M.S. in Electrical Engineering. *Advisor:* Seungwon Shin

RESEARCH INTERESTS

I work on providing security guarantees over an agent's action space: guaranteeing what an agent does even when the model is turned against us. I approach this by making agents act through typed, machine- and human-checkable interfaces. **Maris** provides sound enforcement of usage-control policies for inter-agent communication and tool calls [11], the **Web Verbs** project brings authorization and visibility to web agents through typed browser actions, and ongoing work statically proves properties of the code an agent writes. These three thrusts build on my prior work measuring how deployed agents fail in the wild [1, 2, 10].

HONORS & AWARDS

Future Leaders of AI, ACM AI Leadership Summit *2026*
Distinguished Paper Award, ACM CCS — top ∼1% of accepted papers *2025*
1st Place, UC Berkeley RDI AgentX Competition — Safety track *2025*
2nd Place, UC Berkeley RDI LLM Agents Hackathon — Safety track *2025*
Google Vulnerability Reward Program bounty ($1,337), Gemini CLI [10] *2025*
Internet Society NDSS Symposium Fellowship *2025*
Indiana University Luddy Doctoral Summer Fellowship *2023*
Samsung HumanTech Paper Award (27th edition) *2021*

PUBLICATIONS

(**Jian Cui** = me; * = co-first author. Newest first.)

Refereed Conference Papers

1. "Les Dissonances: Cross-Tool Harvesting and Polluting in Multi-Tool Empowered LLM Agents."
Zichuan Li*, **Jian Cui***, Xiaojing Liao, Luyi Xing. *Network and Distributed System Security Symposium (NDSS'26).*
2nd place, UC Berkeley RDI LLM Agents Hackathon — Safety track.

2. "The Odyssey of `robots.txt` Governance: Measuring Compliance Implications of Web Crawling Bots in LLM Services."
Jian Cui*, Mingming Zha*, XiaoFeng Wang, Xiaojing Liao. *ACM Conference on Computer and Communications Security (CCS'25).*
Distinguished Paper Award — top ∼1% of accepted papers.

3. "Tweezers: A Framework for Security Event Detection via Event Attribution-centric Tweet

Embedding."
Jian Cui, Hanna Kim, Eugene Jang, Dayeon Yim, Kicheol Kim, Jinwoo Chung, Yongjae Lee, Seungwon Shin, Xiaojing Liao. *Network and Distributed System Security Symposium (NDSS'25).*

4. "Malla: Demystifying Real-world Large Language Model Integrated Malicious Services."
   Zilong Lin, **Jian Cui**, XiaoFeng Wang, Xiaojing Liao. *USENIX Security Symposium (USENIX Sec'24).*
   Covered by the *Wall Street Journal*, *Le Monde*, *Fast Company* and *Dark Reading*.
   Top 15, CSAW Best Applied Research Paper Competition.

5. "Ignore Me But Don't Replace Me: Utilizing Non-Linguistic Elements for Pretraining on the Cybersecurity Domain."
   Eugene Jang, **Jian Cui**, Youngjin Jin, Dayeon Yim, Jinwoo Chung, Yongjae Lee, Seungwon Shin. *Findings of NAACL 2024.*

6. "DRAINCLoG: Detecting Rogue Accounts with Illegally-obtained NFTs using Classifiers Learned on Graphs."
   Hanna Kim, **Jian Cui**, Eugene Jang, Chanhee Lee, Yongjae Lee, Jinwoo Chung, Seungwon Shin. *Network and Distributed System Security Symposium (NDSS'24).*

7. "DarkBERT: A Language Model for the Dark Side of the Internet."
   Youngjin Jin, Eugene Jang, **Jian Cui**, Jinwoo Chung, Yongjae Lee, Seungwon Shin. *Annual Meeting of the Association for Computational Linguistics (ACL'23).*

8. "Meta-Path-based Fake News Detection Leveraging Multi-level Social Context Information."
   Jian Cui, Kwanwoo Kim, Seung Ho Na, Seungwon Shin. *ACM International Conference on Information and Knowledge Management (CIKM'22).*

9. "MECaNIC: SmartNIC to Assist URLLC Processing in Multi-Access Edge Computing Platforms."
   Taejune Park, Myoungsung You, **Jian Cui**, Youngjin Jin, Seungwon Shin. *IEEE International Conference on Network Protocols (ICNP'22).*

Under Submission

10. "Measuring and Assessing OAuth Compliance Across the Model Context Protocol Ecosystem."
    Jian Cui, Minsun Shim, Zhou Li, Xiaojing Liao. *Under submission.*
    64 deployed MCP servers tested against 14 security-critical OAuth 2.1 requirements; none fully compliant.
    2 high-severity CVEs, $1,337 Google VRP (`gemini-cli`).

11. "Maris: A Formally Verifiable Privacy Policy Enforcement Paradigm for Multi-Agent Collaboration Systems."
    Jian Cui, Zichuan Li, Luyi Xing, Xiaojing Liao. *Under submission.*
    1st place, UC Berkeley RDI AgentX Competition — Safety track.

12. "ScriptIOC-Bench: A Benchmark for Recognizing Actionable Threat Intelligence from Script-Based Malware using LLMs."
    Hanna Kim, **Jian Cui**, Minkyoo Song, Hwanjo Heo, Seungwon Shin, Kimin Lee, Xiaojing Liao. *Under submission.*

PROFESSIONAL EXPERIENCE

Research Intern, Microsoft Research, Redmond *May – Aug. 2026*

Host: Shuo Chen. Built a new web browsing MCP tool for a product-level agent at Microsoft. It replaces low-level DOM control with vetted, parameterised typed function calls built on Playwright. It provides proper authorization: every action is a named operation, checkable and recorded before it runs. It is also more reliable and more efficient in token cost and latency than raw DOM control.

Applied Scientist Intern, Amazon Web Services *May – Aug. 2024*

Hosts: Pranav Garg, Shweta Garg. Prototyped an availability attack on retrieval-augmented generation: a GCG-style adversarial suffix planted in a retrievable codebase to suppress the model's answers.

Research Intern, S2W Inc., South Korea *Feb. 2022 – Jun. 2023*

Built the dark-web and security-focused corpus and pipelines behind DarkBERT [7], CyberTuned [5]; proposed new anomaly detection methods using graph neural networks [3, 6].

INVITED TALKS

Palo Alto Networks *Aug. 2025*

Towards Trustworthy Agent Development Frameworks: From Attacks to Defenses

AG2/AutoGen Community Talk *Aug. 2025*

A Security-Controlled Development Paradigm for Multi-Agent Collaboration Systems

Guest Lecture, UIUC CS 210: Ethical & Professional Issues *Nov. 2025*

The Odyssey of `robots.txt` Governance

ACADEMIC SERVICE

Program Committee: USENIX Security'27; AsiaCCS'27; NDSS AISCC'24.
Artifact Evaluation Committee: NDSS'25; USENIX Security'24, '25; ACM CCS'24.
Journal & Conference Reviewer: ACL Rolling Review (2025); IEEE TOPS'26; IEEE TIFS'24; Computers & Security (2024).
External Reviewer: ACM CCS'24, '25; IEEE S&P'24, '25; IEEE EuroS&P'24, '25; TheWebConf'24; USENIX VehicleSec'25.
Volunteer: ACM CCS'25.

TEACHING

Teaching Assistant, UIUC CS 562: Advanced Topics in Security, Privacy & ML *Fall 2025*
Teaching Assistant, KAIST TS 251: Data Science Overview *Spring 2020, Spring 2021*